

Class 5 Maths Olympiad — Prep Kit

A practice kit for Class 5 (age 10–11) Mathematics Olympiad exams — SOF-IMO, Silverzone iOM, Unified Council UIMO, and similar school-level olympiads. Companion to [Class-4-Maths-Olympiads](#), built the same way: syllabus and exam pattern researched first, then a week-by-week practice plan, topic worksheets, and full-length sample papers.

Primary target exam

The pattern and worksheets are built against **SOF International Mathematics Olympiad (IMO), Class 5** — the most widely taken olympiad of this type. Class 5's syllabus is a direct step up from Class 4: larger numbers (7–8 digits), plus new topics — percentage, ratio & proportion, and an introduction to profit/loss and simple interest. Topic overlap with Silverzone iOM and Unified Council UIMO is mapped out in [docs/01-syllabus.md](#).

Repository layout

```
docs/
  01-syllabus.md      Full topic-wise syllabus, section by section
  02-exam-pattern.md  Marks, timing, levels, negative marking, cutoffs
  03-practice-plan.md 12-week schedule: what to practice, and when
assignments/
  README.md          Naming convention + full topic list (Weeks 1–11)
  01-number-sense/ ... 11-achievers-hots/ 11 topic worksheets, each with an answer key
  12-achievers-level2/ ... 15-geometry-level2/ Level-2 topic worksheets (harder, for Round-1 qualifiers)
sample-papers/
  README.md          How the full-length papers map to the real pattern
  sample-paper-01.md ... sample-paper-09.md 9 full 50-question papers (IMO pattern, 60 marks, 60 min)
  sample-paper-01-answer-key.md ... sample-paper-09-answer-key.md
                                Papers 1–5: Level-1. Papers 6–9: Level-2 track.
pdf/
  Print-ready PDF of every file above, mirroring the same folder structure
tools/
  generate-pdfs.js    Regenerates pdf/ from the Markdown
12th-phy-derivations/
  README.md          Unrelated bonus content — see note below
```

Note: `12th-phy-derivations/` is unrelated to the Class 5 Maths Olympiad plan above — it's a CBSE Class 12 Physics board-exam derivations checklist, kept here at the user's request rather than in its own repo.

How to use this

1. Read `docs/01-syllabus.md` and `docs/02-exam-pattern.md` once, so you know what "done" looks like.
2. Follow `docs/03-practice-plan.md` week by week.
3. Every assignment and sample paper is self-contained: questions first, answer key (with short explanations) in a separate file.
4. The sample paper is meant to be done **strictly timed** (60 minutes, no calculator) to build real exam pace.

Regenerating PDFs

```
cd tools
npm install
```

```
npx playwright install chromium # one-time, downloads a local Chromium
node generate-pdfs.js
```

It auto-discovers every `.md` file in the repo and titles each PDF from that file's first `# heading` — no code changes needed as content is added.

Status

Complete. Syllabus/pattern research, the 12-week plan, all 11 weekly assignments (Weeks 1–11), and all 9 full-length sample papers (5 Level-1 + a 4-paper Level-2 track for students who qualify Round 1) are built out — each with an answer key, and each arithmetic/logic answer verified programmatically. PDF versions of every file are in `pdf/`. Four dedicated **Level-2 topic assignments** (Achievers/HOTS, Logical Reasoning, Percentage/Ratio/Profit-Loss/SI, Geometry — the areas where Level 2 gets conceptually harder, not just numerically bigger) are also built, in `assignments/12-achievers-level2/` through `15-geometry-level2/`.